

Figure 1: Pig

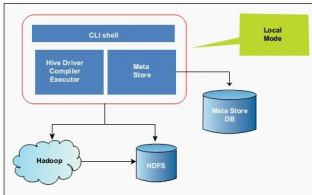


Figure 2: Hive

a primary key to work best, but do not worry if your table structure inherently doesn't have it; it will be created for you but without affecting your table's metadata structure.

HBase

HBase is a distributed column oriented database built on top of HDFS. It is the Hadoop application to use when you need real-time read/write random access to a very large data set. HBase provides APIs that enable development in practically any programming language, and is well suited for sparse data sets. It is written in Java and doesn't enforce relationships within the data. The key point in HBase is that it doesn't care about data types, storing an integer in one row and a string in another, in the same column. It stores a key value pair and stores versioned data. The HBase shell is written with JRuby (the Ruby implementation of JRE) wrapping the Java client API (i.e., one can access Java libraries). HBase runs in three different modes: standalone (running in a single JVM on a single machine), pseudo-distributed (running multiple JVMs on a single machine), and full-distributed (running multiple JVMs on multiple machines).

Zookeeper

This is a distributed coordination service for distributed applications. Problems like centralised configuration, synchronisation, partial failures, deadlocks, race conditions and networking are all solved with Zookeeper. It actually handles the nitty-gritty of distributed application development on the Hadoop ecosystem so that the developer can focus on

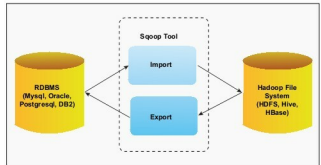


Figure 3: Sqoop

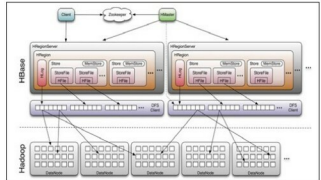


Figure 4: HBase

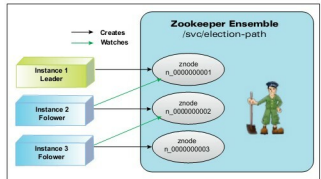


Figure 5: Zookeeper

functionality. Zookeeper always has an odd number of nodes in the cluster as the selection of the master is via voting. Zookeeper has a leader, follower and observer. In the leader, the writes are quorum based and are committed by followers. The followers forward the writes to the leader. Only one leader can write and commit to a file, and all the requests come to the leader via the followers. In case the leader goes down, voting is done among followers to select the leader. Observers just observe the results of votes but never take part in the voting process. **END**

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